

Aurora Cell: Technical Whitepaper

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1. Introduction

Veltrana Energy AG, based in Freiburg, develops the Aurora Cell, a sodium-ion battery for home energy storage. The project started in 2023 under chief engineer Dr. Tobias Hallmann. The first pilot line in Offenburg produced 4,800 cells in 2025.

2. Chemistry

The Aurora Cell uses a Prussian white cathode and a hard carbon anode. It contains no lithium, cobalt or nickel. The nominal cell voltage is 3.1 V and the gravimetric energy density is 158 Wh/kg, compared with roughly 250 Wh/kg for a typical lithium iron phosphate cell of the same format.

3. Performance

In cycling tests at 25 degrees Celsius, the Aurora Cell retained 91 percent of its capacity after 4,000 full cycles. At minus 20 degrees Celsius it still delivered 86 percent of its rated capacity, which is its main advantage over lithium-based cells for unheated garages and outdoor installations.

4. Test Results

Table 1 summarizes results from the Offenburg test lab (March 2026).

Test	Condition	Result
Cycle life	25 C, 1C/1C	91% capacity after 4,000 cycles
Cold performance	-20 C	86% of rated capacity
Fast charge	10% to 80%	14 minutes
Nail penetration	Full charge	No fire, max 62 C
Self-discharge	30 days, 25 C	2.1% loss

5. Cost and Outlook

Veltrana estimates a cell cost of 61 euros per kWh at a production volume of 2 GWh per year, about 20 percent below comparable lithium iron phosphate cells. A second factory in Kehl is planned to open in 2028 with an annual capacity of 5 GWh. The main open risk is the supply of hard carbon, which currently comes from a single supplier, Karbonit Werke.

6. Contact

Questions about this whitepaper can be sent to the fictional Veltrana research office, attention Dr. Lea Brandt.